CHALMERS

CANDIDATE

DAT232-0038-UPZ_2026-10-29

TEST

DAT232 HT26 LP1 Ordinarie tentamen - J

Subject code	--
Evaluation type	1
Test opening time	29.10.2026 08:30
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1- Instructions

Question	Status	Marks	Question type
ℹ			Information or resources

Sektion 2 - Multiple Choice

Question	Status	Marks	Question type
ℹ			Information or resources
1.1	Wrong	0/1	Multiple Choice
1.2	Correct	1/1	Multiple Choice
1.3	Correct	1/1	Multiple Choice
1.4	Wrong	0/1	Multiple Choice
1.5	Correct	1/1	Multiple Choice
1.6	Wrong	0/1	Multiple Choice
1.7	Correct	1/1	Multiple Choice
1.8	Wrong	0/1	Multiple Choice
1.9	Correct	1/1	Multiple Choice
1.10	Correct	1/1	Multiple Choice
1.11	Correct	1/1	Multiple Choice
1.12	Wrong	0/1	Multiple Choice
1.13	Correct	1/1	Multiple Choice
1.14	Wrong	0/1	Multiple Choice
1.15	Wrong	0/1	Multiple Choice
1.16	Correct	1/1	Multiple Choice

1.17	Correct	1/1	Multiple Choice
1.18	Correct	1/1	Multiple Choice
1.19	Correct	1/1	Multiple Choice
1.20	Wrong	0/1	Multiple Choice
1.21	Correct	1/1	Multiple Choice
1.22	Correct	1/1	Multiple Choice
1.23	Correct	1/1	Multiple Choice
1.24	Correct	1/1	Multiple Choice
1.25	Wrong	0/1	Multiple Choice
1.26	Correct	1/1	Multiple Choice
1.27	Wrong	0/1	Multiple Choice
1.28	Correct	1/1	Multiple Choice
1.29	Correct	1/1	Multiple Choice
1.30	Correct	1/1	Multiple Choice

Sektion 3 - Interactive Task

Question	Status	Marks	Question type
i			Information or resources
2.1	Correct	10/10	Inline Gap Match

Sektion 4 - Essay questions

Question	Status	Marks	Question type
i			Information or resources

3.1	Answered	Manually marked	Essay
3.2	Answered	Manually marked	Essay
3.3	Answered	Manually marked	Essay
3.4	Answered	Manually marked	Essay

1.1

Proposition	Reason
According to the IEEE definition of a requirement from 1990, a capability that a system must possess to satisfy a formally imposed standard can constitute a requirement.	The definition recognises such a capability as a requirement only if an individual user has also explicitly requested it.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✔
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✖

Maximum marks: 1

1.2

Proposition	Reason
Developing suitable requirements can require combining knowledge from domain experts and software developers.	Domain experts may understand the work but lack knowledge of technical possibilities, while developers may understand technical possibilities but lack knowledge of the work.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Maximum marks: 1

1.3

Proposition	Reason
Observing a stakeholder performing their work can reveal requirements that were not mentioned during an interview.	Stakeholders may perform familiar routines without consciously describing them, while observation can expose those routines and the information or support they require.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Maximum marks: 1

1.4

Proposition	Reason
According to Glinz's classification, a constraint on a system is a non-functional requirement.	In this classification, every non-functional requirement is an attribute of the system.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✔
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✖

Maximum marks: 1

1.5

Proposition	Reason
Recording the rationale for a requirement can help future maintainers understand why the system was built in a particular way.	The rationale preserves the purpose or justification behind the requirement for people who were not involved in the original decision.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✔
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Maximum marks: 1

1.6

Proposition	Reason
According to Lauesen's quality criteria, a requirements specification is complete when all necessary requirements are included.	According to the same criteria, a requirement is verifiable when it is possible to determine whether it has been met.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition. ✓
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✗

Maximum marks: 1

1.7

Proposition	Reason
Filling in every section of a requirements specification template does not, by itself, establish that all necessary requirements have been included.	A necessary requirement may concern an aspect of the system that the chosen template does not cover.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✓
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Maximum marks: 1

1.8

Proposition	Reason
If every requirement in a specification reflects a stakeholder need, the specification necessarily includes all necessary requirements.	According to Lauesen's quality criteria, correctness means that each documented requirement reflects a need.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✔
- E) Both the proposition and the reason are false. ✖

Maximum marks: 1

1.9

Proposition	Reason
The level of detail needed for a good enough requirements specification can be determined independently of the development context.	In a small agile project, face-to-face communication can compensate for some detail omitted from written requirements.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✔
- E) Both the proposition and the reason are false.

Maximum marks: 1

1.10

Proposition	Reason
Context diagrams are useful for defining the scope of a product.	A context diagram shows the product boundary and the product's relationships with entities outside that boundary.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✓
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Maximum marks: 1

1.11

Proposition	Reason
Data expressions are suitable for specifying the structure of messages in a communication protocol.	Data expressions represent information as simplified screens containing realistic example data.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✓
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Maximum marks: 1

1.12

Proposition	Reason
Writing each requirement as a separate “the system shall” statement ensures that the requirements specification is unambiguous.	A feature requirement can omit information needed to determine how the system should behave.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✔
- E) Both the proposition and the reason are false. ✖

Maximum marks: 1

1.13

Proposition	Reason
Assumption busting can help a requirements workshop discover possibilities that participants initially excluded.	The technique involves listing assumptions about the problem and challenging whether those assumptions are correct.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✔
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Maximum marks: 1

1.14

Proposition	Reason
In the BrightSparks technique, participants use famous-person personas to stimulate ideas.	The technique asks participants to generate new ideas by randomly pairing and combining previously generated ideas.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✔
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✖

Maximum marks: 1

1.15

Proposition	Reason
Combinational creativity generates new ideas by combining existing ideas.	Combinational creativity is defined by changing the rules that determine the space of possible ideas.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✖
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✔
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Maximum marks: 1

1.16

Proposition	Reason
Convergent thinking should be excluded from Creative Problem Solving because evaluating ideas conflicts with creativity.	Deferring judgement during idea generation is a core principle of Creative Problem Solving.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✓
- E) Both the proposition and the reason are false.

Maximum marks: 1

1.17

Proposition	Reason
Creativity techniques should replace stakeholder elicitation when the aim is to discover novel requirements.	Stakeholders can contribute only requirements they have already seen implemented in existing systems.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✓

Maximum marks: 1

1.18

Proposition	Reason
An ordinal prioritisation scale expresses how many times more important one requirement is than another.	Ranking requirement A above requirement B establishes that A is twice as important as B.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✓

Maximum marks: 1

1.19

Proposition	Reason
Assigning requirements independently to high, medium, and low priority groups can leave a team with insufficient guidance about what to exclude from a release.	Independent grouping does not force requirements to be compared against each other and can result in too many requirements being classified as high priority.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✓
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Maximum marks: 1

1.20

Proposition	Reason
According to Ramesh and Edwards' definitions, tracing from requirements to tests is horizontal traceability.	In these definitions, vertical traceability connects artifacts of the same type.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason do not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✔

Maximum marks: 1

1.21

Proposition	Reason
Pre-requirements traceability can support validation of requirements.	Linking requirements to their sources and originating business goals helps reviewers assess whether the requirements address real needs.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Maximum marks: 1

1.22

Proposition	Reason
Requirements validation checks whether documented requirements match stakeholders' needs.	Checking that a specification is internally consistent is sufficient to establish that its requirements match stakeholders' needs.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✓
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Maximum marks: 1

1.23

Proposition	Reason
Separating a compound requirement into individual obligations can make their fulfilment easier to assess independently.	Separate obligations allow reviewers to record which obligations are met when only part of the originally combined requirement is fulfilled.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✓
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Maximum marks: 1

1.24

Proposition	Reason
Once a formal model has been proved to satisfy every checked property, validating it against stakeholders' needs is unnecessary.	Formal verification establishes stated properties of a model under specified assumptions.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✓
- E) Both the proposition and the reason are false.

Maximum marks: 1

1.25

Proposition	Reason
Agreeing on a checklist before reviewers individually inspect a specification can help ensure that the planned quality checks are covered.	In perspective-based reading, reviewers examine a specification from a particular role's perspective, such as that of a tester.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason do✓ not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✗
- E) Both the proposition and the reason are false.

Maximum marks: 1

1.26

Proposition	Reason
A CRUD matrix can support verification by exposing possible inconsistencies between data requirements and task descriptions.	A CRUD matrix establishes whether each requirement contributes to a stakeholder's business goal.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✓
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Maximum marks: 1

1.27

Proposition	Reason
Confirmation is part of a user story and describes how satisfactory implementation will be assessed.	The INVEST criteria include that user stories should be small and estimable.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason do not explain the proposition. ✓
- C) The proposition is a true statement, but the reason is false. ✗
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Maximum marks: 1

1.28

Proposition	Reason
In large-scale agile development, cross-cutting quality requirements can be managed adequately by each team independently, without coordination between teams.	A quality requirement can affect several components, functional requirements, and teams.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✓
- E) Both the proposition and the reason are false.

Maximum marks: 1

1.29

Proposition	Reason
Reducing ambiguity in requirements can help reduce the risk of unfair treatment of stakeholders.	Ambiguous requirements can permit different interpretations that overlook some stakeholders' needs or lead to biased decisions.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✓
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Maximum marks: 1

1.30

Proposition	Reason
Informed consent during requirements elicitation requires participants to understand the purpose and expected use of their contributions and to participate voluntarily.	Obtaining informed consent removes the ethical need to limit data collection to information necessary for the requirements work.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✓
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Maximum marks: 1

2.1 For each situation, select the technique and explanation that together most directly address the stated information need. Select one technique and one explanation per situation. Options may be reused, and some options may not be used.

Each correct technique gives **1 point**, and each correct explanation gives **1 point**. The two selections are scored independently. Incorrect or missing selections give **0 points**.

Help

Collects comparable responses	CRUD check
Formal proof	Defines database structure
Estimates implementation effort	Prototype evaluation
Allows assessing interactions	Facilitated workshop
Uncovers objective	Questionnaire
Interview (ask why)	Checks internal consistency
Reveals tacit knowledge	Enables joint negotiation
Observation	Generates novel feature ideas

1. Staff struggle to explain familiar routines, and previous interviews missed important exceptions.
   - Technique: Observation ✓
   - Explanation: Reveals tacit knowledge ✓
2. Several stakeholder groups disagree about how a shared process should work.
   - Technique: Facilitated workshop ✓
   - Explanation: Enables joint negotiation ✓
3. The team needs comparable responses from several hundred geographically dispersed users.
   - Technique: Questionnaire ✓
   - Explanation: Collects comparable responses ✓
4. A stakeholder requests a specific interface feature, but its underlying purpose is unclear.
   - Technique: Interview (ask why) ✓
   - Explanation: Uncovers objective ✓
5. Users cannot judge whether a proposed interaction supports their work from its textual description.
   - Technique: Prototype evaluation ✓
   - Explanation: Allows assessing interactions ✓

Maximum marks: 10

3.1 Topic: Managing Quality Requirements and Negotiating Trade-offs (10p)

Write an essay explaining how requirements engineers can make quality expectations concrete and negotiate an appropriate level of quality. Consider a software product with limited development resources and stakeholders who place different importance on its quality characteristics.

Include all of the following concepts:

- Quality grid: identifying and communicating the importance of quality factors
- Operationalisation using Planguage: measurement scale, measurement procedure, and acceptance threshold
- Open metric or open target: explain and apply at least one, including the associated customer risk
- Trade-offs between quality requirements and their relationship to functional requirements
- Negotiation and prioritisation of competing quality expectations

Use one coherent software-product example. Include one concrete, assessable quality requirement and one justified trade-off. Explain how uncertainty and stakeholder differences would be handled before accepting a proposed solution.

Your answer will be graded on how accurately you explain and apply the concepts, and how clearly you connect them in your argument.

Maximum length: 750 words.

Write your answer in the box below. Changes are saved automatically.

Let's say we have a course registration platform for students. Students want to register courses easily. However, it is not clear in what kind of level easy means to users, and it is not verifiable for the developers, so it needs an agreed criterion.

Let's use Planguage with scale - what to measure, meter - how to measure, and goal - acceptance threshold:
scale - system's response time.
meter - take 500 concurrent users doing course registration in a testable environment, and measure the time it takes for the system to respond with status change successfully from "not registered" to "registered".
goal - the system's response time of registration confirmation with status change should be within 2 seconds.

And this is an open metric and open target that software engineers have came up so it can be verifiable, however, it needs further discussion with customers with workshop to see if it can be an acceptance criterion, because requirements validation checks whether documented requirements match stakeholders' needs.

Also, stakeholders can have conflict requirements as well. For example, students want to have more steps, and administrators want to have less steps. Seems the requirements from them are contradicting, but software engineers can ask them the reasons behind their need, then they clarified that students want to have confirmation message to avoid miss click or withdraw unwanted actions, and administrators want to have automatic approval to save time. Then the negotiation can be set an agreed deadline for registration, and all registrations happen before deadline are automatically approval after students confirmed registration, and students can cancel the registrations with confirmation as well. And after the deadline, if students want to make changes, they need to send a request for administrators' approval. In this case, software engineers figure out that there is a midway because they understand what stakeholders actually need.

Words: 305

Maximum marks: 10

3.2 Topic: Traceability for Change Impact Analysis and Quality Assurance (10p)

Write an essay explaining how traceability can support the analysis and quality assurance of a requirements change. Consider a change to a requirement that may affect other requirements, design or code, and tests.

Include all of the following concepts:

- Purpose-driven traceability and relevant artifacts
- Traceability Information Model: artifact types and allowed or required link types
- Change impact analysis: candidate impact set and actual impact set
- Traceability maintenance: missing, incorrect, and stale links
- Using traceability to support requirements verification and validation

Use one coherent software-system change to illustrate your argument. Explain which links would be useful, how you would investigate the proposed impact set, and how you would check both the changed requirements and the trace information.

Your answer will be graded on how accurately you explain and apply the concepts, and how clearly you connect them in your argument.

Maximum length: 750 words.

Write your answer in the box below. Changes are saved automatically.

Requirements can be linked to business goals or other requirements. For example, one use case claims that user clicks the course registration button, and the request is sent to the system, the system changes the status of selected course from 'not registered' to 'registered'. This feature can also be investigated as quality requirement: the system's response time of registration confirmation with status change should be within 2 seconds.

For example, if a data dictionary field gets changed due to database migration or other reasons, but the development and testing team are not in sync with that, then if that data was used to be retrieved and displayed in the software, now it would show error, and the testing would also fail before production. Therefore, using an authoritative rule and referencing it in the remaining content in the documentation can improve modifiability, and reduce the risk of documentation inconsistency.

Using traceability also support requirements verification and validation. For example, during a CRUD matrix, the data shows update but not create, developers can further investigate where the data comes from, whether it is an external import or a missing data creation functionality or a missing description from stakeholder due to tacit knowledge. So the candidate impact set can be all three assumptions, but after investigation, the actual impact set would be clear.

Words: 220

Maximum marks: 10

3.3 Topic: Requirements Documentation in an Agile Development Context (10p)

Write an essay explaining how a suitable combination of requirements documentation and communication practices can support agile development. Consider a product developed by several agile teams, with shared functionality and quality requirements that affect more than one team.

Include all of the following concepts:

- Combining documentation approaches: explain and combine at least two of structured, feature-based, model-based, and scenario-based approaches
- Breadth-first and just-in-time requirements engineering
- User stories: card, conversation, and confirmation
- Coordination of requirements across teams and levels of responsibility
- Trade-offs between documentation effort, communication, and the risks of missing or outdated requirements

Use one coherent software-product example to illustrate your argument. Justify what should be documented, when it should be elaborated, and how it should be kept useful as the product changes.

Your answer will be graded on how accurately you explain and apply the concepts, and how clearly you connect them in your argument.

Maximum length: 750 words.

Write your answer in the box below. Changes are saved automatically.

In agile development, combination of using requirements documentation and communication would help different teams stay sync on the same level.

The documentation is usually structured with different sections based on business goals, requirement levels and requirement types. Each category has its own focus, for example, user stories use card, conversation, and confirmation, where card describes the role, capability, and purpose; conversation checks the understanding; confirmation checks whether it meets stakeholders' need. However, user case focuses on the operations between user and system, for example, in user story "As a student, I want to register courses.", and in user case "User clicks the course registration button, and the request is sent to the system, the system changes the status of selected course from 'not registered' to 'registered'.". In this case, developers clearly know what functionalities need to be implemented.

Also, nowadays, when several agile teams work with shared functionality and quality requirements together, each team still might be responsible for different but related parts. One works with development, one works with testing, one works with database, is a common structure. Therefore, keep efficient communication and consistency in the documentation is more important than we think. For example, if a data dictionary field gets changed due to database migration or other reasons, but the development and testing team are not in sync with that, then if that data was used to be retrieved and displayed in the software, now it would show error, and the testing would also fail before production.

Therefore, using an authoritative rule and referencing it in the remaining content in the documentation can improve modifiability, and reduce the risk of documentation inconsistency. Even though it seems that communication and this documentation maintenance takes a lot of time effort, but without doing this, the payoff can be way higher than expected.

Words: 302

Maximum marks: 10

3.4 Topic: Managing Ethical Concerns in Requirements Engineering (10p)

Write an essay explaining how requirements engineering can help identify and address ethical concerns, while also ensuring that requirements engineering activities are conducted ethically.
Include all of the following concepts:

- Affected stakeholders, including overlooked or underrepresented groups
- Ethical principles: explain and apply at least two of autonomy, beneficence, nonmaleficence, justice, and explicability
- Ethics in requirements elicitation and analysis
- Translating ethical concerns into concrete requirements
- Ethical trade-offs and lifecycle review

Use one coherent software-system example to illustrate your argument. Explain how your proposed responses could be assessed and revisited.

Your answer will be graded on how accurately you explain and apply the concepts, and how clearly you connect them in your argument.

Maximum length: 750 words.

Write your answer in the box below. Changes are saved automatically.

In the requirement engineering, it is extremely important to understand the stakeholders' needs. And how to understand what their requirements are to minimize the chances of missing overlooked or underrepresented groups.

There are elicitation techniques to use: interview, observation, and questionnaire. Usually, it is good to start with interview and observation first to get a sufficient domain knowledge, then use questionnaire to see how widespread with the specified areas. Since these techniques involve quite a lot of participants, to protect participants' privacy and credentials, it is important whether they know that the information are going to be used in the future. Most common way is to use anonymous questionnaire with informed consent for participants to choose that they understand the purpose and expected use of their contributions and to participate voluntarily, since nowadays surveys can simply be shared digitally with links, so if the participant chooses disagree with the informed concent, then that questionnaire should not be taken. This can also be translated into concrete requirements in "stakeholders who participated during the elicitation were informed and agreed with the information to be used".

However, an ethical trade-offs can be that, first, it is hard to track
with anonymous participants if there are further investigations needed, which limits the potentialities of software maintenance after releasing. Second, it is also hard for anonymous participants to withdraw their participation. However, even interviews or observations that usually saved participants contact information, and they decide to withdraw their participation can also cause trade-off, such as attrition bias, as if the ones who quit are all screen reader users, and the ones left are usual laptop users, and the registration platform is not functional on the screen reader, but since these participants have left, then the testing result would should that 100% of the participants successfully to use the registration platform, which is untrue.

Words: 308

Maximum marks: 10